

Engineering Economics (CSE-481)

Week 1 — BIST 100 Financial Analytics with Python & MCP AI Agents

A simple tutorial: market data → indicators → support/resistance → backtest → AI agent

BIST 100 + Yahoo Finance
Python + MCP

Dr. Alper ÖZCAN
Akdeniz University

- Understand how BIST stocks are represented as time-series data.
- Download BIST stock prices with Python and Yahoo Finance.
- Read Open, High, Low, Close and Volume (OHLCV) data.
- Explain the logic of support/resistance and pivot points.
- Understand how an MCP-based AI agent can call financial tools.

DATA → RULES → SIGNALS

SIGNALS → BACKTEST

TOOLS → AI AGENT

The course moves from simple rules to explainable, tool-using financial agents.

Course Roadmap

Updated for BIST, Python and AI Agents

1 BIST 100 & Yahoo Finance

2 Algorithmic Trading

3 Technical Indicators

4 Backtesting Strategies

5 Time Series with Pandas

6 Lab: BIST Data

7 Lab: Support/Resistance

8 Midterm

9 Engineering Economics
Concepts

10 Fundamental Analysis &
Valuation

11 Money Management

12 MCP AI Agent

13 Project Presentations

14 Project Presentations

Grading (Undergraduate)



Project requirements: Proposal + final report • Groups of two students are allowed

Python combines financial data, indicators and backtesting into one analysis pipeline.

What is BIST 100?

- BIST 100 is a benchmark index of major companies traded on Borsa İstanbul.
- Each company has a time series: price, volume and other market variables.
- Our goal is not to guess prices manually — it is to design repeatable algorithms.

Example BIST symbols

ASELS.IS

THYAO.IS

TUPRS.IS

BIMAS.IS

GARAN.IS

EREGL.IS

Yahoo Finance typically uses the “.IS” suffix for Borsa İstanbul equities.

Our Data Sources

Yahoo Finance

Historical prices
OHLCV
Index data

Fintables

Financial statements
Valuation ratios
Company
fundamentals

TCMB / TÜİK

Interest rates
Inflation
FX and macro
context

Python combines these sources into one analysis pipeline.

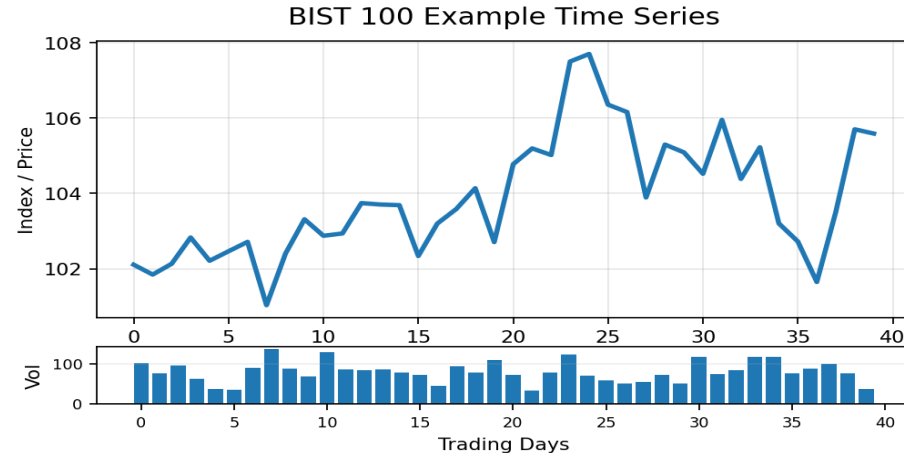
What is BIST 100?

- BIST 100 is a benchmark index of major companies traded on Borsa İstanbul.
- Each company has a time series: price, volume and other market variables.

Our Data Sources

Yahoo Finance / BIST 100

Historical prices, OHLCV and index data



Fintables

Financial statements, valuation ratios and company fundamentals

Özet Rapor
Şirket Bilgileri
Akış
Karne
Sermaye Artırımları ve Temettüleri
Faaliyet Verileri
Takas Analizi
Fon Pozisyonları
Rakip Analizi
Analist Tavsiyeleri
Yeni İş İlişkileri
✓ Finansal Tablolar
Bilanço
Gelir Tablosu
Nakit Akım Tablosu
✓ Oran Analizi
Rasyo Analiz Tablosu
Piyasa Çarpanları
Likidite Oranları

finansal zeka evo cevaplasın.

Aselsan Elektronik Sanayi ve Ticaret A.Ş. (ASELS) Bilanço Analizi

Bilanço Analizi sayfamızda şirketin açıkladığı bilançolarını Türk Lirası, dolar ve euro bazında görebilirsiniz. Şirketin döviz bazında satışlarını, döviz bazında karını, dolar bazlı satışlarını, euro bazlı satışlarını analiz edebilirsiniz.

Düzeltilmiş Tüm Periyotlar TRY % Excel'e Aktar

Bu sayfadaki finansal tablolar 2026/6 dönemi enflasyon endeksi ile düzeltilmiştir. Orijinal tablolara "Düzeltilmemiş Tablolar" filtresi ile ulaşabilirsiniz. [Detaylı bilgi](#)

Bilanço Kalemleri	2026/6 ₺	2026/3 ₺	2025/12 ₺	2025/9 ₺	2025/6 ₺
Dönen Varlıklar					Bir TRY
Nakit ve Nakit Benzerleri	39.468.926	33.180.286	34.251.653	17.742.125	20.862.613
Finansal Yatırımlar					
Ticari Alacaklar	47.167.326	44.436.221	45.760.320	35.517.588	34.694.067
Diğer Alacaklar	10.135.228	7.258.331	6.015.164	4.711.529	3.782.655
Stoklar	106.485.325	91.756.385	86.741.403	80.483.578	69.963.223
Peşin Ödenmiş Giderler	37.804.766	37.721.441	26.434.952	26.783.591	25.280.782
Diğer Dönen Varlıklar	8.503.994	7.887.511	6.854.238	6.239.100	5.908.505
Toplam Dönen Varlıklar	248.765.565	222.240.175	206.057.730	171.477.510	160.491.844

Python combines market data and financial data into one analysis pipeline.

```
pip install yfinance pandas matplotlib  
  
import yfinance as yf  
import pandas as pd  
import matplotlib.pyplot as plt
```

1. Install packages

2. Import libraries

3. Download BIST data

For class work, use a virtual environment or a clean notebook environment.

Example 1 — Download ASELS Data

```
import yfinance as yf

df = yf.download(
    "ASELS.IS",
    start="2024-01-01",
    end="2026-01-01"
)

print(df.head())
```

Typical output

Date	Open	High	Low	Close	Volume
2025-09-01	164.0	168.5	162.7	167.9	...
2025-09-02	168.0	171.2	165.4	169.6	...

One row = one trading day

Understanding OHLCV

Open

First traded price

High

Highest price

Low

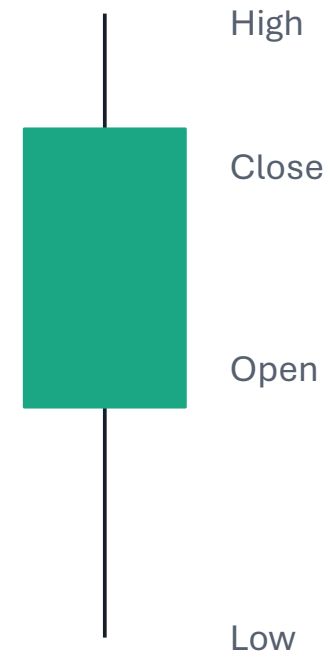
Lowest price

Close

Last traded price

Volume

Trading activity



Green candle: Close > Open

Support and Resistance



- **Support:** a price zone where buyers appear.
- **Resistance:** a price zone where sellers appear.
- Price may bounce, break, or retest these levels.

Support = Buy Zone?



Breakout



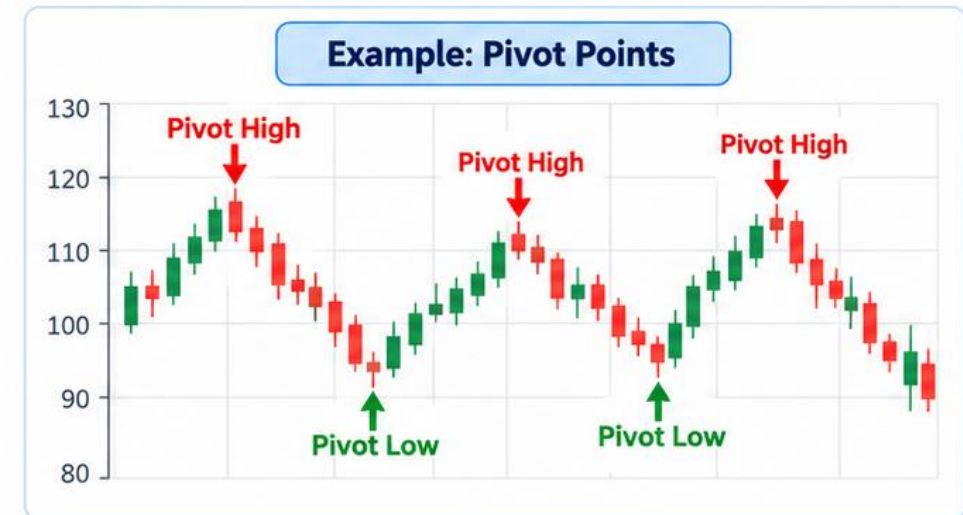
Resistance = Sell Zone?



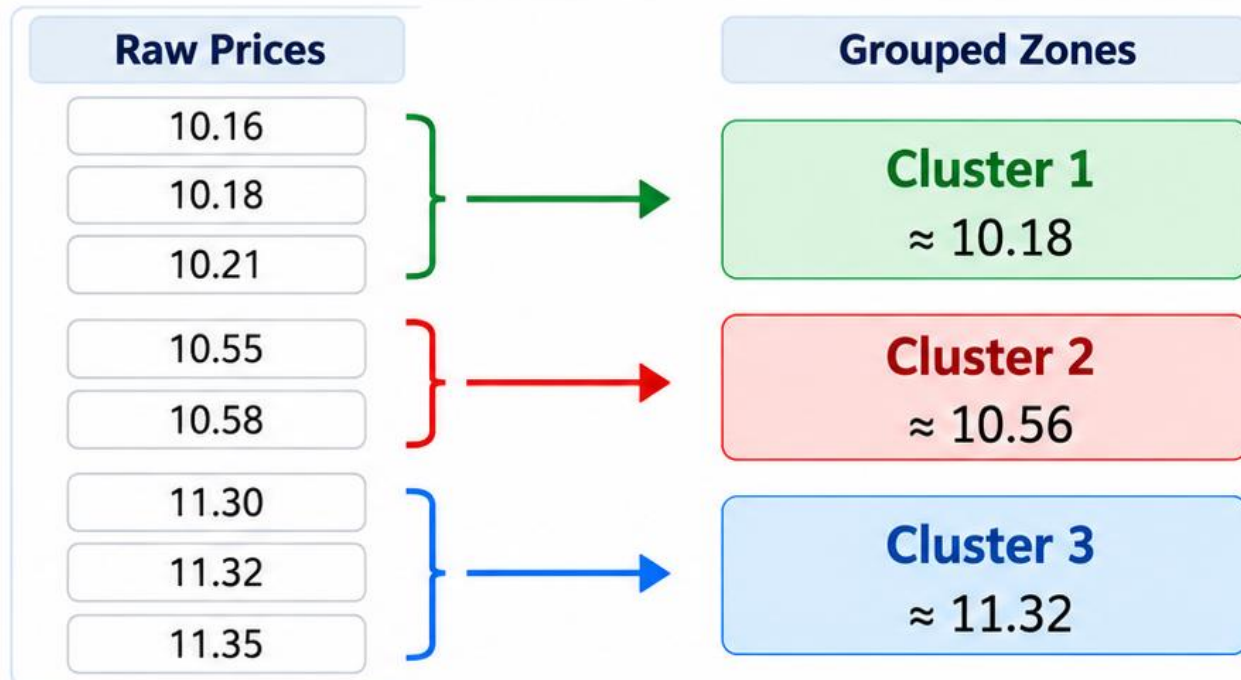
How Does the Algorithm Find the Lines?



- Local highs suggest resistance.
- Local lows suggest support.
- Repeated reactions make a level stronger.



Group Nearby Prices



- Markets react to zones, not exact points.
- Nearby prices can belong to the same level.
- Use a tolerance such as 1%–2%.



Strength Score for a Level

Strength Score Formula

Strength Score =

Number of Touches

+

Recency Weight

+

Volume Weight

+

Reversal Strength

Example Strength Scores

Level	Touches	Score
10.18	5	7.2
10.55	2	2.8
11.02	4	6.1
11.32	3	4.7

More Touches = Stronger Level



More touches = stronger level

- Repeated reactions increase importance.
- Recent, high-volume reactions may carry more weight.

Technical Indicators – Start Simple

SMA

(Simple Moving Average)



Trend direction

EMA

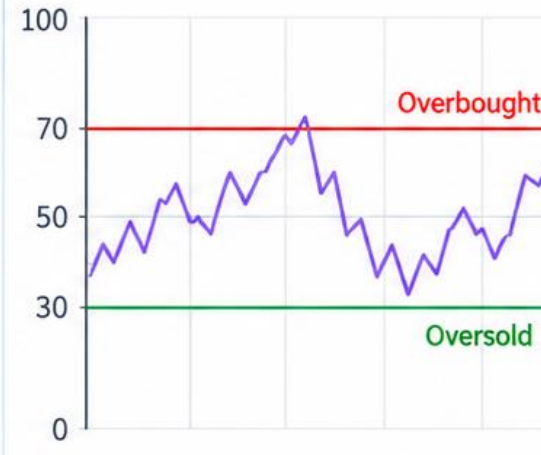
(Exponential Moving Average)



Faster trend signal

RSI

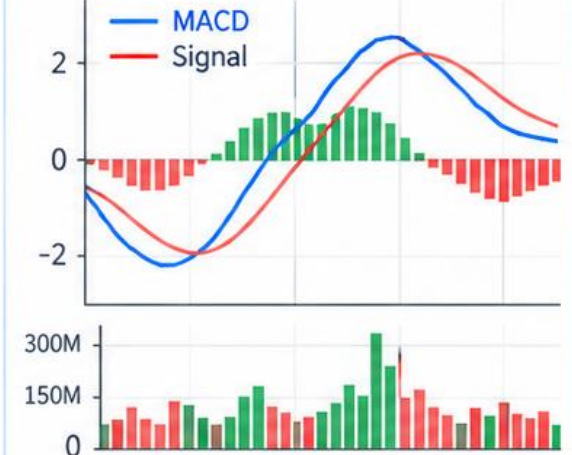
(Relative Strength Index)



Overbought / Oversold

MACD + Volume

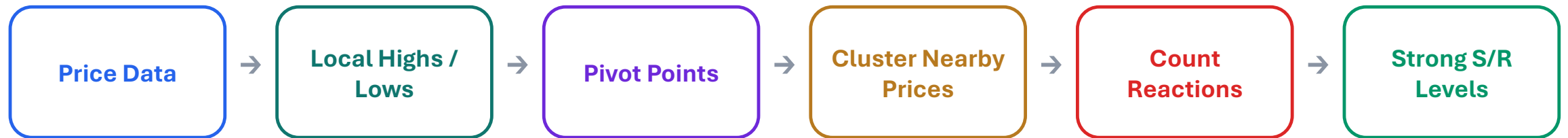
(Momentum and Activity)



Momentum and activity

- Start with a few indicators.
- Do not use too many signals at once.
- Combine indicators with support / resistance.

How Does the Algorithm Find the Lines?



Important idea: support/resistance is usually a zone, not one exact price.

Example 2 — Detect Pivot Highs and Lows

```
window = 5

df["PivotHigh"] = (
    df["High"] == df["High"]
    .rolling(2*window+1, center=True).max()
)

df["PivotLow"] = (
    df["Low"] == df["Low"]
    .rolling(2*window+1, center=True).min()
)
```

Interpretation

Pivot High → Resistance?

Pivot Low → Support?

The “?” matters: one pivot is only a candidate. Repeated reactions make a level stronger.

Example 3 — Group Nearby Prices

Observed pivot prices

10.16

10.18

10.21

These are close enough to represent one zone.



≈ 10.18 SUPPORT

```
tolerance = 0.015    # 1.5%
```

```
distance = abs(price - level) / level
```

```
if distance < tolerance:  
    same_zone = True
```

Clustering reduces many noisy pivot prices into a small number of meaningful levels.

Strength Score for a Level

Not every line is equally important.

- Number of touches
- Recent reactions
- High-volume reactions
- Size of the reversal

```
score = (  
    touches * 1.0  
    + recent_touches * 1.5  
    + high_volume_reactions * 1.2  
    + reversal_strength * 1.0  
)
```

**Keep only the strongest
levels.**

Technical Indicators — Start Simple

SMA

Trend

RSI

Momentum

MACD

Trend + Momentum

Volume

Participation

```
df["SMA20"] = df["Close"].rolling(20).mean()  
df["SMA50"] = df["Close"].rolling(50).mean()
```

From Indicator to Trading Rule

Example rule

SMA20 > SMA50 → BUY

SMA20 < SMA50 → SELL

```
df["Signal"] = 0
```

```
df.loc[df["SMA20"] > df["SMA50"], "Signal"] =  
1
```

```
df.loc[df["SMA20"] < df["SMA50"], "Signal"] =  
-1
```

Rule-based ≠ guaranteed profit. We must test it.

```
df["Return"] = df["Close"].pct_change()

df["Strategy_Return"] = (
    df["Signal"].shift(1) * df["Return"]
)

equity = (1 + df["Strategy_Return"]).cumprod()
```

Compare

Strategy Return

vs.

BIST 100 Return

Avoid look-ahead bias: use only information that would have been available at that time.

Fundamental Analysis with Fintables

- Revenue and revenue growth
- EBITDA and operating profitability
- Net profit
- Debt and liquidity
- P/E, P/B, EV/EBITDA, ROE

Example Fundamental Score

```
score = 0
if revenue_growth > 0.20: score += 1
if roe > 0.20:             score += 1
if net_debt_ebitda < 2:    score += 1
if pe < sector_pe:         score += 1
```

The course combines market data (technical) with company data (fundamental).

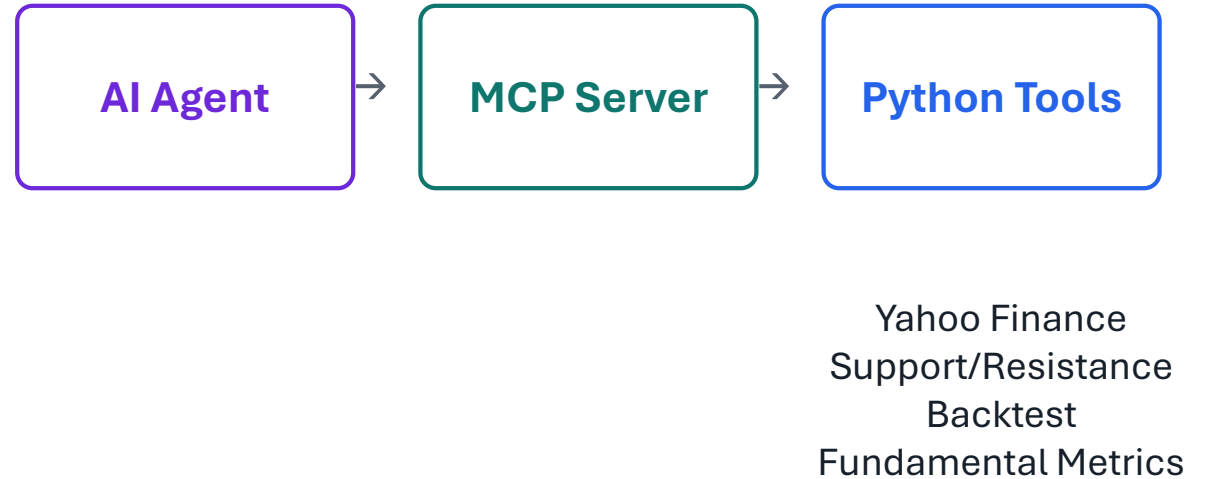
MCP AI Agent — What Changes?

Without an agent

User manually runs scripts and interprets outputs.

With an MCP agent

The AI can discover and call tools, then explain the result.



Example MCP Tools for the Project

```
get_stock_history(symbol, start, end)
```

Download OHLCV data

```
find_support_resistance(symbol)
```

Return strongest price zones

```
calculate_indicators(symbol)
```

SMA, RSI, MACD, volume

```
run_backtest(symbol, strategy)
```

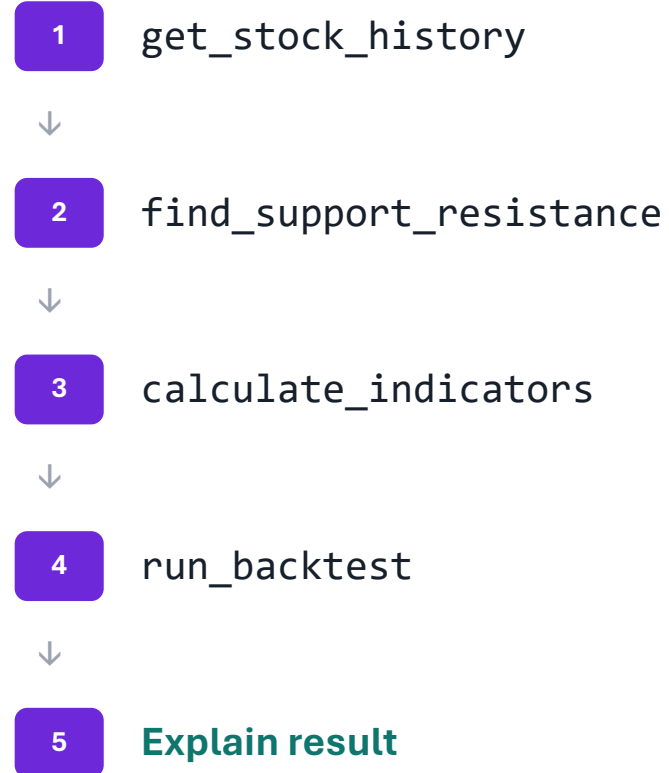
Evaluate strategy performance

```
get_fundamental_metrics(symbol)
```

Read key company metrics

AI Agent Example

User: “Analyze ASELS.IS. Find support/resistance and compare the strategy with BIST 100.”



The LLM decides which tool to call; the financial calculations remain in deterministic Python functions.

Week 1 Mini Lab

Task

- Choose one BIST stock (e.g., ASELS.IS).
- Download two years of daily data.
- Print the first five rows and explain OHLCV.
- Plot the closing price.
- Identify at least two possible support/resistance zones.
- Write one simple trading rule in plain English.

Deliverable

- Python notebook
- 1 chart
- 3–5 sentences of interpretation
- No real-money trading required

Semester Project — Overview

BIST 100 Stock Analysis, Prediction and AI-Agent Assisted Strategy Development



Goal: build an explainable research prototype — not an investment recommendation system.

Semester Project — Phases

1. Architecture & Strategy Development

Define data pipeline, stock universe, indicators, rules and MCP tools.

2. Backtesting & Simulation

Run historical simulations and compare with a benchmark.

3. Prediction & Evaluation

Optional ML model; evaluate accuracy and financial performance.

4. Final AI-Agent Demo

Ask natural-language questions and execute approved analysis tools.

Project Evaluation & Expected Outcomes

- Clean Python implementation and reproducible data pipeline.
- At least one rule-based trading strategy.
- Backtest results with benchmark comparison.
- Clear evaluation metrics and discussion of limitations.
- Simple MCP AI agent that uses project tools safely.
- Final report + presentation + source code.

Suggested Metrics

Cumulative Return

Max Drawdown

Sharpe Ratio

Hit Rate / Accuracy

Benchmark Difference

AI Use Policy

- AI tools may support idea generation, coding assistance and understanding.
- Students remain responsible for the analysis, interpretation and final work.
- Prompts and materially used AI outputs should be documented in the appendix.
- Financial outputs must be verified against the underlying data and code.

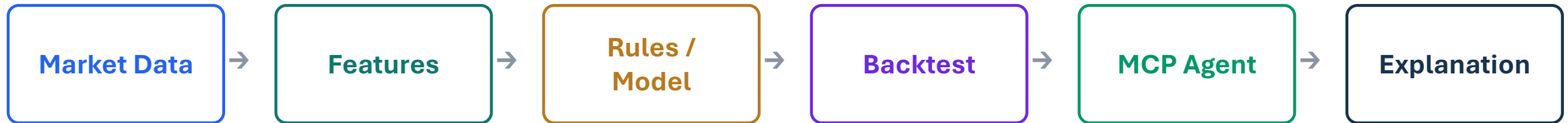
AI ASSISTS

**Python
CALCULATES**

Student EXPLAINS

Takeaway

A financial analysis system is a pipeline.



Next week: Algorithmic Trading — from financial intuition to explicit rules.

Project Scope: Technical + Fundamental Analysis

Calculate technical indicators such as SMA, EMA, RSI, MACD and volume-based indicators.

Automatically detect support and resistance zones using pivot points and clustering.

Use Fintables to examine the selected company's income statement, balance sheet and cash-flow information.

Calculate and interpret fundamental indicators such as Revenue Growth, EBITDA Growth, Net Profit Growth, ROE, P/E, P/B and Net Debt/EBITDA.

Create a simple fundamental score and combine it with technical-analysis signals.

Develop at least one rule-based trading strategy.

Backtest the strategy without look-ahead bias.

Focus: BIST 100 + Yahoo Finance + Fintables + Python + MCP AI Agent

Fundamental Analysis with Fintables

Category	Example Metrics	Question to Answer
Growth	Revenue Growth, EBITDA Growth, Net Profit Growth	Is the company expanding and improving earnings?
Profitability	ROE, ROA, EBITDA Margin, Net Profit Margin	How efficiently does the company generate profit?
Leverage	Net Debt/EBITDA, Debt/Equity	How risky is the debt structure?
Liquidity	Current Ratio, Quick Ratio	Can the company meet short-term obligations?
Valuation	P/E, P/B, EV/EBITDA	How expensive is the company relative to earnings, equity or operating profit?

Students should interpret these metrics together rather than using a single ratio alone.

Simple Fundamental Score

Example scoring rules:

- +1 Revenue Growth > 15%
- +1 EBITDA (Earnings Before Interest, Taxes, Depreciation and Amortization) Growth > 15%
- +1 Net Profit Growth > 15%
- +1 ROE (Return on Equity) > 20%
- +1 Net Debt / EBITDA < 2
- +1 P/E (Price-to-Earnings Ratio) below sector benchmark

Fundamental Score = 0 ... 6

0–2 Weak

3–4 Neutral

5–6 Strong

This score can be merged with technical signals to create a combined ranking or trading system.

Simple Fundamental Score

Scoring Rules (1 point each)

- **+1** Revenue Growth > 15%
- **+1** EBITDA Growth > 15%
- **+1** Net Profit Growth > 15%
- **+1** ROE > 20%
- **+1** Net Debt / EBITDA < 2
- **+1** P/E below sector average

Fundamental Score = 0 ... 6

Score Interpretation

0-2 | Weak

3-4 | Neutral

5-6 | Strong



This score can be combined with technical signals for stock ranking or trading.

Combined Strategy and Backtesting

Technical Analysis: SMA, EMA, RSI, MACD, Volume, Support/Resistance

Fundamental Analysis: Revenue, EBITDA, Net Profit, ROE, P/E, P/B, Net Debt/EBITDA

Optional Macro Data: USD/TRY, interest rate, inflation

Combine these into a final rule-based BUY / HOLD / SELL decision

Run backtesting on historical data and compare with Buy-and-Hold and XU100

Avoid look-ahead bias: only use information available at that date

$$\text{Combined Score} = 0.4 \times \text{Technical Score} + 0.4 \times \text{Fundamental Score} + 0.2 \times \text{Macro Score}$$

Suggested extension: expose technical/fundamental functions as MCP tools for an AI agent.